CHAPTER 6

APPLICATION OF DERIVATIVES

CHAPTER ESSENCE AT A GLANCE:
Rate of change of quantities, increasing and decreasing function tests, and finding local and absolute maxima and minima.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Instantaneous Rates of Change:** The first derivative represents the instantaneous rate of change of a physical or geometric quantity with respect to another variable, such as expanding area or leaking volume with respect to time.

• **Monotonicity and Derivative Signs:** A differentiable function is strictly increasing on an interval where its first derivative is strictly positive, and strictly decreasing where its first derivative is strictly negative.

• **Extreme Values and Critical Points:** Local extrema occur exclusively at critical points where the first derivative vanishes or fails to exist, with concavity revealed by the sign of the second derivative.

• **Geometric Tangents and Normals:** The first derivative evaluated at a point yields the slope of the tangent line to the curve, while the normal line possesses a perpendicular slope equal to the negative reciprocal of the tangent slope.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\frac{dA}{dt} = 2\pi r\frac{dr}{dt}, \qquad \frac{dV}{dt} = 4\pi r^2\frac{dr}{dt}$$

Time rates of change for circular area and spherical volume

$$f'(x) > 0 \to \text{Increasing}, \qquad f'(x) < 0 \to \text{Decreasing}$$

First derivative test for monotonic intervals

$$f'(c) = 0 \text{ and } f''(c) < 0 \to \text{Max}; \qquad f''(c) > 0 \to \text{Min}$$

Second derivative test for local extrema

$$y - y_0 = m(x - x_0), \qquad m_{\text{tangent}} = \frac{dy}{dx}\Big|_P, \qquad m_{\text{normal}} = -\frac{1}{m_{\text{tangent}}}$$

Equations and slopes of tangent and normal lines at point P

$$x^2 + y^2 = L^2 \to \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}$$

Rate of descent along vertical wall in sliding ladder model

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Strictly vs Non-Strict] Strictly increasing requires strictly positive f'(x) > 0 and open interval (a, b). Do not include endpoints where derivative is zero.

• **Alert:** [Trap Alert: Second Derivative Sign] Negative second derivative f''(c) < 0 means MAXIMUM (concave down), while positive f''(c) > 0 means MINIMUM! Do not invert.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Find the rate of change of area of a circle with respect to radius r when r = 6 cm. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Rate of Change of Area]** Area of circle: A = πr^2. Rate of change with respect to radius r: dA by dr = d by dr(πr^2) = $2\pi r$. [1 Mark]

• When r = 6 cm: dA by dr = 2π (6) = 12π cm^2 per cm = 37.7 cm^2 per cm. [1 Mark]

• **[Boxed Result: dA by dr = 12π cm^2 per cm]**

2. Prove that the function f(x) = cos x is strictly decreasing in (0, π) and strictly increasing in (π, 2π). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Derivative of cos x]** f(x) = cos x $\Rightarrow$ f'(x) = - sin x. [0.5 Mark]

• 1. In interval (0, π): sin x > 0. Therefore, f'(x) = - sin x < 0. Since f'(x) < 0, f(x) = cos x is STRICTLY DECREASING in (0, π). [1 Mark]

• 2. In interval (π, 2π): sin x < 0. Therefore, f'(x) = - sin x = - (negative) > 0. Since f'(x) > 0, f(x) = cos x is STRICTLY INCREASING in (π, 2π). [0.5 Mark]

• **[Boxed Result: Strictly decreasing in (0, π); Strictly increasing in (π, 2π)]**

3. The length x of a rectangle is decreasing at 3 cm/min and width y is increasing at 2 cm/min. When x = 10 cm, y = 6 cm, find rates of change of perimeter and area. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Given Data]** dx by dt = -3 cm/min (decreasing), dy by dt = +2 cm/min (increasing). At instant x = 10 cm, y = 6 cm. [0.5 Mark]

• **[Rate of Change of Perimeter P = 2(x + y)]** dP over dt = 2 (dx by dt + dy by dt) = 2 (-3 + 2) = 2 (-1) = - 2 cm/min (perimeter is decreasing). [1 Mark]

• **[Rate of Change of Area A = x y]** dA by dt = x (dy by dt) + y (dx by dt) = 10(2) + 6(-3) = 20 - 18 = + 2 cm^2/min (area is increasing). [1.5 Marks]

• **[Boxed Result: Perimeter decreases at 2 cm/min; Area increases at 2 cm^2/min]**

4. Show that the function f(x) = $x^3 - 3x^2 + 3x - 100$ is increasing on R. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Derivative Analysis]** f(x) = $x^3 - 3x^2 + 3x - 100$. f'(x) = $3x^2 - 6x + 3 = 3(x^2 - 2x + 1) = 3(x - 1)^2$. [2 Marks]

• Since $(x - 1)^2 \geq 0$ for all x R: f'(x) = $3(x - 1)^2 \geq 0$ for all x R. f'(x) vanishes only at the isolated point x = 1 (does not vanish on any open interval). Therefore, f(x) is strictly INCREASING on the entire set R. [1 Mark]

• **[Boxed Result: f'(x) = $3(x - 1)^2 \geq 0 \Rightarrow$ f is increasing on R]**

5. Find all points of local maxima and local minima of f(x) = $2x^3 - 6x^2 + 6x + 5$. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[First & Second Derivatives]** f(x) = $2x^3 - 6x^2 + 6x + 5$. f'(x) = $6x^2 - 12x + 6 = 6(x^2 - 2x + 1) = 6(x - 1)^2$. Setting f'(x) = 0 gives x = 1 as the only critical point. [1.5 Marks]

• **[First Derivative Test]** For x < 1: f'(x) = $6(x - 1)^2 > 0$. For x > 1: f'(x) = $6(x - 1)^2 > 0$. f'(x) does not change sign as x increases through 1 (it is positive on both sides). Therefore, x = 1 is NEITHER a point of local maximum NOR local minimum (it is a Point of Inflection). There are NO local extrema! [1.5 Marks]

• **[Boxed Result: x = 1 is a Point of Inflection; No points of local maxima or minima]**

6. A ladder 5 m long is leaning against a wall. The bottom is pulled along ground at 2 m/s. How fast is height on wall decreasing when foot is 4 m from wall? [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Setup & Pythagorean Relation]** Ladder length 5 m: $x^2 + y^2 = 5^2 = 25$. Given dx by dt = +2 m/s. When foot is x = 4 m from wall: $y = \sqrt{(25 - 4^2)} = \sqrt{(25 - 16)} = \sqrt{9} = 3$ m. [1.5 Marks]

• **[Differentiation with respect to time t]** d by dt$(x^2 + y^2)$ = d by dt(25) ⇒ 2x (dx by dt) + 2y (dy by dt) = 0. x (dx by dt) + y (dy by dt) = 0 ⇒ 4(2) + 3 (dy by dt) = 0 ⇒ 8 + 3 (dy by dt) = 0. dy by dt = - 8 over 3 m/s. The height on the wall is decreasing at the rate of 8 over 3 m/s (2.67 m/s). [2.5 Marks]

• **[Boxed Result: Height decreases at 8 over 3 m/s]**

7. Find the intervals in which the function $f(x) = 2x^3 - 3x^2 - 36x + 7$ is strictly increasing or strictly decreasing. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Derivative & Critical Points]** $f(x) = 2x^3 - 3x^2 - 36x + 7$. $f'(x) = 6x^2 - 6x - 36 = 6(x^2 - x - 6) = 6(x - 3)(x + 2)$. Set f'(x) = 0 ⇒ x = 3, x = -2. These points divide real line into three disjoint intervals: (-∞, -2), (-2, 3), (3, ∞). [2 Marks]

• **[Interval Sign Analysis]** 1. Interval (-∞, -2): Choose x = -3: f'(-3) = 6(-6)(-1) = +36 > 0 ⇒ Strictly Increasing. 2. Interval (-2, 3): Choose x = 0: f'(0) = 6(-3)(2) = -36 < 0 ⇒ Strictly Decreasing. 3. Interval (3, ∞): Choose x = 4: f'(4) = 6(1)(6) = +36 > 0 ⇒ Strictly Increasing. [2 Marks]

• **[Boxed Result: Strictly Increasing on (-∞, -2) ∪ (3, ∞); Strictly Decreasing on (-2, 3)]**